

Amine Idelhaj

DATA SCIENTIST | MACHINE LEARNING ENGINEER

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Profile

Data Scientist with a Master's degree in Data and Intelligent Systems, specialized in Machine Learning, Data Analytics, and NLP. Experienced in developing data-driven solutions, predictive models, and analytical PoCs using Python and modern ML frameworks.

EXPERIENCE

Data Scientist & Generative AI Engineer – QSE/RSE SaaS Platform, Ian Clic – Lille, France Mai 2025 – March 2026

- Collaborated with stakeholders to identify business needs and define AI-driven use cases for document intelligence.
- Developed Python APIs to integrate Generative AI capabilities into a SaaS platform for QHSE/CSR document management.
- Designed and implemented a Retrieval-Augmented Generation (RAG) system powered by Mistral AI for intelligent document analysis and question answering.
- Built semantic search pipelines to retrieve and process relevant information from audit and compliance documents.
- Contributed to the development of AI-powered solutions for document understanding and knowledge extraction.

Data science Intern, Greyc – Caen, Normandie, France Mars 2024 – Sept 2024

- Developed and fine-tuned a **DistilBERT** model for cyberattack classification from textual cybersecurity data.
- Built NLP preprocessing pipelines including text cleaning, tokenization, feature extraction, and dataset preparation.
- Trained and evaluated Transformer-based models using PyTorch, Hugging Face Transformers, and Scikit-Learn.
- Optimized model performance through hyperparameter tuning and comparative evaluation of classification approaches.
- Assessed model effectiveness using Precision, Recall, F1-Score, and Confusion Matrix metrics.
- Presented research findings and model performance analyses to academic supervisors and research teams.

AI Engineer Intern, Elden – Casablanca, Morocco Févr 2023 – Oct 2023

- Developed AI-based features for secure document management.
- Implemented document classification and anomaly detection solutions.
- Contributed to blockchain-based data integrity and traceability mechanisms.

PROJECTS

MERN Blog Application Jan 2025

- Developed a full-stack blog application using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Implemented user authentication, CRUD functionalities, and secure database management.
- Designed responsive interfaces and RESTful APIs for content management.
- Technologies: React.js, Node.js, Express.js, MongoDB.

Music Recommendation System Dec 2023

- Developed a recommendation system based on user listening behavior and preferences.
- Performed data preprocessing, feature engineering, and recommendation analysis.
- Improved recommendation relevance through machine learning techniques.
- Technologies: Python, Pandas, Scikit-Learn.

AddicTest - Addiction Risk Prediction Platform Dec 2023

- Developed a Django-based web application to predict addiction risk using Machine Learning models.
- Performed exploratory data analysis on a dataset of more than 1,800 participants, including demographic,

- psychological and substance use data.
- Built data visualization dashboards and correlation analyses to identify addiction risk factors.
- Technologies: Python, Django, Machine Learning, Pandas, Matplotlib, Data Analysis.

Pneumonia Detection using VGG19

Dec 2023

- Built a deep learning model for pneumonia detection from chest X-ray images.
- Applied transfer learning using the VGG19 architecture to improve classification accuracy.
- Evaluated model performance using standard medical image classification metrics.
- Technologies: TensorFlow, Keras, Computer Vision, Deep Learning.

Stock Market Analysis using Spark Streaming

Nov 2023

- Processed and analyzed real-time stock market data using Apache Spark Streaming.
- Developed distributed data processing workflows for financial analytics.
- Generated insights from streaming data and market trend analysis.
- Technologies: Apache Spark, Python, Big Data Analytics.

Online Exam Monitoring System

Oct 2023

- Developed a computer vision-based online proctoring system to detect cheating behaviors during remote examinations.
- Implemented face detection, eye tracking, gaze estimation, and head pose analysis using machine learning and computer vision techniques.
- Designed a multi-student monitoring solution capable of tracking participants simultaneously through webcam streams.
- Technologies: Python, OpenCV, Computer Vision, Machine Learning, Eye Tracking.

EDUCATION

Master – Data and Intelligent Systems , Université Claude Bernard Lyon 1	Sept 2023 – Sept 2024
Engineering Degree – Digital Data Engineering & AI , ENSAM Rabat	Sept 2020 – Août 2023
CPGE First Prépa Méknes , Mathematics & Physics	Sept 2018 – Juin 2020

Technical Skills

- Machine Learning:** Scikit-Learn, TensorFlow, PyTorch, NLP.
- Programming Languages:** Python, SQL, Scala, JavaScript, C/C++.
- Big Data:** Apache Spark, Apache Kafka, Hadoop, ETL, Distributed Data Processing.
- Data Analytics:** EDA, Data Mining, Forecasting, KPI Development, Reporting, ETL/ELT Pipelines, Business Intelligence.
- Generative AI:** LLMs, Prompt Engineering, AI Workflow Automation, OpenAI APIs.
- Python Libraries:** Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow, PyTorch.
- Databases:** PostgreSQL, MySQL, Oracle, MongoDB, Data Warehousing.
- Visualization:** Power BI, Matplotlib, Seaborn, Advanced Microsoft Excel.
- Cloud & Deployment:** Docker, Kubernetes, CI/CD, AWS, OVHcloud, GCP, Amazon S3, EC2.

Languages

Arabic (Native) French (Fluent) English (Fluent)